



Measuring Gentrification and Displacement in Greater London

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[Paper first received, August 1998; in final form, March 1999]

Summary. The use of aggregate data to understand the linkages between gentrification and displacement has been considered problematic because of a lack of refinement or closeness to the nature of these processes as well as lacking the ability to ‘track’ displacees. This paper presents the results of pioneering work designed to overcome these problems by combining cross-sectional census data with spatially re-aggregated longitudinal census data (the Longitudinal Study). Using established approaches to measure gentrification, via proxy measures, and devising others for its potential displacees, the work demonstrates a displacement effect clustered around gentrified wards. Attempts are made to quantify flows of displacement relative to city-wide changes over the decade. The paper concludes that, although replacement and displacement are difficult to distinguish, displacement appears prevalent for certain groups and this requires further research initiatives to explore a process that is socially and psychologically harmful.

Introduction

Gentrification refers to a process of class succession and displacement in areas broadly characterised by working-class and unskilled households first identified in the East End of London by Glass (1964). Smith and Williams (1986, p. 1) define gentrification as

the rehabilitation of working-class and derelict housing and the consequent transformation of an area into a middle-class neighbourhood.

Smith and LeFaivre (1984, p. 60) have argued that “capitalism is based precisely on its ability to displace the working class in all sorts of situations” and history has shown us many examples of the ways in which legislatures, the wealthy and the politically influential have managed to move the poor when it was profitable or expedient to do so.

The clearance of the Scottish Highlands (Prebble, 1969), the development of the railways (Kellett, 1979), slum clearance programmes (Young and Willmott, 1957), housing grants (McCarthy, 1974; Hamnett, 1973) all moved large sections of the population, inevitably the poorest. What these processes have in common is that they take a line of least resistance. They attack the tenure insecurity of those with the quietest political voice and with least equity, in order that compensation may be set to a correspondingly low level, ignoring the effects that such movement incurs on individuals and wider social networks (Le Gates and Hartman, 1986; Chan, 1986).

While some have acknowledged that displacement from gentrification has been both prevalent and socially harmful (notably,

Sumka, 1979; Hartman, 1979; Le Gates and Hartman, 1986; Marcuse, 1986), little research has been conducted in Britain to measure its extent. This despite the fact that extensive gentrification has occurred over the past 30 years or so in areas like London. Perhaps the most obvious reason for this research gap is that it is very difficult to track displacees. It is much easier to research the more tangible manifestation of gentrification, given that it is an observable end-state predicated on the removal or voluntary migration of previous residents.

This apparent lack of data provides the basis for this paper and the beginning of a search to uncover the nature and extent of gentrification-induced displacement. Of course, gentrification may not necessarily displace anyone at all; people may migrate by choice rather than by force or economic necessity. However, even the apparently benign upgrading of previously vacant property may bring a 'price shadowing' of nearby rents and property prices which may create pressures on those with fewer resources (Hall and Ogden, 1992). While cities in competition, and the depopulation of the metropolis, create concern for its future (Rogers, 1999), it is pertinent that we should ask at what cost should revitalisation occur?

Gentrification appears to improve the physical and social fabric of an area, while appealing to the activities of profit-making investment, in run-down areas. Market logic and the ontological status of property relations (Blomley, 1997) appear to dictate that desirable central-city space should be used only by those with more money. Such arguments are one side of a coin; the flipside is that social problems are evacuated through the 'improvement' of neighbourhoods and are thereby used as evidence that gentrification has positive impacts on social problems. Such a view is giving way to an awareness that it is diversity and not homogeneous environments, be they rich or poor, which are most successful at reducing neighbourhood problems (Page and Boughton, 1997; Atkinson and Kintrea, 1998).

This paper provides evidence to suggest,

first, that gentrification was widespread and, in many cases, comprehensive; and, secondly that, while causality cannot be definitive, it is probable that working-class household displacement was a reality for a significant number of households during the 1980s.

Gentrification and Displacement in the US

Gentrification has been dealt with by a number of writers, all seeking to understand a process that stands in contrast with received wisdom on the nature of neighbourhood location (Hamnett and Randolph, 1984). Significant contributions (London and Palen, 1984; Smith and Williams, 1986; van Weesep and Musterd, 1991) have grappled with its theoretical and empirical dimensions in a number of countries, though often with an arguably global city focus. This paper does not seek to engage with these debates as such. Instead, the aim is to take received wisdom on the conceptualisation of gentrification, to operationalise it, measure it and see if this is associated with a displacement effect.

An extensive literature on gentrification-induced displacement exists in North America where quantification has been more successful. This provides us with a template to explore the manifestation of these processes in London. Displacement is a problematic subject given inevitable political wrangling over the adequacy of data and debates about what constitutes displacement itself (Lee and Hodge, 1984). Displacement occurs in circumstances where

any household is forced to move from its residence by conditions which affect the dwelling or its immediate surroundings (US Department of Housing and Urban Development, in Le Gates and Hartman, 1981, p. 214).

Different displacement types have implications for the operationalisation and measurement of displacement. Marcuse (1986), for example, identifies four types:

- (1) economic/physical (where residents are priced out of a dwelling through rent increases or by physical means);

- (2) last-resident displacement (only the last resident is counted as displaced);
- (3) chain displacement (counting includes the number of residents over time who have been displaced from a property); and
- (4) exclusionary displacement (a figure which includes those who have been unable to access property because it has been gentrified).

For the purposes of the research, it should be understood that the measurement used was of last-resident displacement since, though longitudinal data were used, census data only give snapshots of location at 10 year intervals so that intervening moves cannot be measured.

Levels of displacement have been a contentious issue. In the US, Sumka (1979) estimated that 500 000 households, roughly 2 million people, were displaced annually. Le Gates and Hartman (1981 and 1986) viewed this as a purposeful undercount by the Department of Housing and Urban Development. Le Gates and Hartman estimated that at least a further half million people being displaced annually would be both an "approximate and conservative measure".

Separating gentrification and displacement out from wider processes of social change, incumbent upgrading, voluntary migration and welfare and labour market changes provides complex problems for measuring such processes. Further, it is exceedingly hard to distinguish between gentrification as a form of neighbourhood replacement or displacement.

Displacement in London

Although London has been the focus of increasing polarisation and occupational restructuring (Hamnett, 1976; Hall and Ogden, 1992; Harloe, 1992), there has been little work done on displacement. Only two British studies stand out in this respect. The Department of the Environment carried out a survey in 12 inner London boroughs (McCarthy, 1974) to find out whether existing residents

were benefiting from house renovation and, if not, to see what happened. Differences between outward-moving households and in-moving ones (based around the hypothesised involvement of gentrification) were also measured.

The study found that household movement prior to renovation was significant and noted that "the improvement of living conditions did not benefit the original residents" (McCarthy, 1974, p. 3). In total, 68 per cent of renovation applications sampled had been preceded by the outward movement of at least one household and in total almost three-quarters of all households had moved away. Of those leaving, 80 per cent were tenants, as might be expected. A sequence of vacation, sale and then improvement appeared prevalent. By far the most common reason for moves was landlord harassment (43 per cent). The process was described as one in which the

housing costs associated with improved (and improvable) dwellings in inner London ... tend to act as a social sieve (McCarthy, 1974, p. 19).

Work by Lyons (1995) used the longitudinal study (hereafter LS) at a borough level to examine the effect of increasing polarisation and professionalisation on potential displacee groups in London over the census period 1971–81, looking at the socioeconomic, geographical and migratory aspects of the process. She found that local migration was associated with low-status households, while longer-range migration may be associated with those of a higher status, indicating a relationship to constraint and choice respectively. For Lyons, displacement was linked to gentrification and consumer choice for the gentrifiers. However, for displacees, because of their lack of market resources, constraint and coercion characterised such moves.

Overview of the Approach Taken

For this study, work began with the use of standard cross-sectional 1981 and 1991 census data in order to get a picture of social

change across London at electoral ward level. Proxy indicators were used for gentrification and displacement and were correlated to see if increasing professionalisation, as a proxy for gentrification, was negatively related to proxies for those involved in displacement. Following this, four new borough-sized areas were constructed from those wards which had experienced above-average levels of professionalisation, forming four new areas that ranged from low to high levels of gentrification. These new areas (labelled 'G' areas) were used as the locus for exploring changes based on migration and static status changes.

After establishing that migration was significantly greater than internal status changes over the period, analysis shifted to the significance of exit flows from the four new 'G' areas. Finally, comparisons were made with the rest of London to see if the moves in the 'G' areas were more pronounced, as one would expect, than in London as a whole.

The selection of proxy variables for gentrification and displacement was largely based on the preceding displacement and gentrification literature. Boundary changes of wards were dealt with using a computer package designed to approximate 1991 boundaries to those of 1981. However, caution should be used in an unquestioned acceptance of the results as error-free. The study of gentrification, often described at a neighbourhood level, lends itself to an analysis pegged at the smallest level of census analysis, the enumeration district (ED; having an approximate mean value of 200 households or roughly 500 persons). Moves made over short distances may not be picked up by analysis using larger spatial units. However, analysis at ED level is not possible because boundary changes can lead to total mismatches between 1981 and 1991. In addition, the use of data on socioeconomic group uses 10 per cent counts which means that sampling errors may be pronounced at the ED level (Openshaw, 1995).

Approaches using census data to measure and map gentrification have been adopted in

the past (Hamnett and Williams, 1979; Galster and Peacock, 1986) using proxy measures of gentrification based on spatially bounded increases of professionals and managers. Dangschat (1991) found that, even in existing areas which have been gentrified, the continuation of the process may displace ever higher income groups (Dangschat, 1991). Therefore, all wards were considered open to gentrification regardless of the number of professionals at the baseline period.

Research on gentrification-induced displacement in London may be informed by the work of academics and government research, predominantly in the US. In the research subsequently described, proxy indicators were used to establish the existence of gentrification and displacement. Gentrification proxies were based on an operationalisation of the definition of this concept as increases in professionals and managers while, for displacement, the proxy is decreases in the numbers of a set of vulnerable groups.

Measuring Gentrification and Displacement

Each variable was taken as a percentage of its total relevant population (for example, total professionals as a percentage of the working population, 16 years old and more) to derive percentage point changes to measure social change. These were then weighted by population size changes over the period. Socioeconomic groups (SEGs) used to construct a 'gentrifying class'—as used by Lyons (1995)—were adopted as a measurement which minimised the possibility of measurement error and comparability problems between each census.

Previous work (for example, Hamnett and Williams, 1979) has used increases in head of households in this category. This is a notoriously male view of labour and one ignorant of the, now acknowledged, female contribution to gentrification (Warde, 1991) and to the professional class in general (Davies, 1996). Phillips (1993) has addition-

ally argued that one of the distinctions between rural and urban examples of gentrification is that they are symmetrical (dual earner/couples) and asymmetrical (single) household structures respectively, further highlighting the contribution of females.

The group selected comprised socioeconomic groups (SEGs); 1.1, 1.2, 2.1, 2.2, 3, 4, 5.1 and 13 (employers in large establishments, managers in large establishments, employers in small establishments, managers in small establishments, professional workers—self-employed, professional workers—employees, ancillary workers and artists and employer/manager farmers). Of course, lower SEGs may have similar abilities to displace those with lower resources than themselves and, in this sense, there is a degree of relativity to the process of displacement. This point has already been noted by Lyons (1995) where junior non-manual workers were held to be potential displacers of lower groups.

Some writers (for example, Hall and Ogdén, 1992) have noted that gentrifiers are often young professionals. However, this study did not attempt an age breakdown, partly because of the complications this would impose when looking at changes over time, but also because other literature shows that older groups may act both as gentrifiers (Parsons, 1980) and as 'ultra-gentrifiers' (Dangschat, 1991) who are able to displace earlier waves of gentrifiers.

Gentrification was indicated by a rate of occupational change in any single ward exceeding the mean rate of increase for London as a whole. This led to the exclusion of all wards with a growth rate of less than 5 percentage points—the city-wide mean for the ward growth of professionals and managers. While professionals and managers were fewer in some wards, they also formed a relatively larger group due to greater losses of other occupational groupings. These were eliminated leaving 133 wards (out of a total of 755 wards in the Greater London area, excluding the City of London). The increase in these remaining wards was in the range of 5–22 percentage points.

Seven unidimensional displacement variables were derived from the literature, as follows:

- (1) *Working class* (SEGs 6, 7, 8, 9, 10 and 15): Junior non-manual workers, personal service workers, foremen and supervisors—manual, skilled manual workers, semi-skilled manual workers and agricultural workers respectively.
- (2) *Unskilled labour*: (SEG 11).
- (3) *Households privately renting*: Renting has often declined in favour of owning and has been associated with a turnover, or displacement, of households (Hamnett and Randolph, 1984; Murie, 1991). Increasing tenure insecurity for such groups has made them even more susceptible.
- (4) *Ethnicity*: Essentially all non-white groups, although this is accepted as crude. Since the census has only used the definition of ethnicity since 1991, a measure of a person's country of birth was used as a proxy. (This measure included those who were born in the New Commonwealth and Pakistan in 1981 and simply the New Commonwealth, which included Pakistan, in 1991).
- (5) *Unemployed*: To achieve broadly comparable figures for 1981, it was necessary to take a figure of those 'seeking work' and 'temporarily seeking' as a basic figure of unemployment; however, in 1991, students who were unemployed were counted in the basic 'unemployed' figure so that their exclusion in 1981 meant that they had to be subtracted to achieve comparability in the 1991 figures.
- (6) *Elderly*: A figure was taken of the percentage of total population in each ward aged 60 years and over.
- (7) *Lone parent*: A figure was taken that included both sexes as heads of a household with children up to the age of 16 as a percentage of the total number of households. It was not possible to exclude professional lone-parent households.

Table 1. Mean percentage point change for each gentrification and displacement variable by quartile

Variable	Top quartile	Second quartile	Third quartile	Lowest quartile
Professionals	15.10	11.06	8.93	7.26
Working class	-17.69	-15.84	-14.70	-13.23
Renting	-6.12	-6.12	-3.75	-3.24
Elderly	-3.04	-2.28	-1.71	-1.75
Unskilled	-3.01	-1.82	-1.23	-1.19
Ethnic minority	-0.66	-0.06	0.54	1.19
Lone parent	3.91	2.94	3.96	3.54
Unemployed	12.59	10.88	11.82	11.58

Sources: 1981 and 1991 Censuses Data.

Descriptive Analysis

Table 1, shows a clear correspondence between rises in each of the gentrification categories for each area—becoming more pronounced in inner London—and decreases in the displacee variables for London as a whole, again becoming stronger for the inner area. This seemed to be taking place in each of the areas with the exception of unemployment and lone parents (later shown to be positively correlated with increases in professionals).

This appears to show an apparent inverse relationship between the gentrification and displacee variables, except for the lone parents and unemployed. The dramatic decreases in working class could be for a number of reasons. First, fewer people are now employed in these occupations after extensive occupational restructuring and the loss of much industrial employment in the central city. Secondly, members of these occupational groups may have migrated from the capital. Finally, part of the impact of increasing numbers of professionals may be a displacement of working-class groups. In reality, it is likely that a combination of these factors is responsible. Certainly, considerable ward variation existed within each quartile, hiding many changes.

There were more 'gentrified' wards in outer (79) than in inner (51) London showing that, in general, the wards with more than average growth were expanding more in the outer than inner areas of London. However, a

roughly similar proportion of ward areas of inner and outer London had been gentrified (17.1 per cent and 17.2 per cent respectively, expressed as a percentage of the wards in each area) though the extent of gentrification in inner London was still, on average, higher than that of outer London.

The Longitudinal Study (LS)

The LS itself allows the life-course of a 1 per cent sample of individuals and their families from the census to be tracked over time, in both their status and geographical location. As a random sample, it is very reliable in its replication of the distributions to be found in the larger census and key events such as cancers, deaths and migrations are also added to the sample.

Moves from one borough to another may appear more significant for the displacee but, equally, may hide moves made only over small distances (Lyons, 1995). This problem is also evident in the LS which preserves anonymity of the 1 per cent sample by allowing disaggregation only to borough level. In order to overcome these limitations, a geography of gentrification was used to build the 'G' areas and this meant that these new areas were not contiguous. This increased the likelihood that moves might be made to another area rather than continuing to be hidden as moves within the same borough.

LS data are restricted to England and Wales so that flows to or from Scotland or migratory moves abroad appear as exits from

the data-set. There is also no information about the sample population during the intervening period so that one cannot tell how many moves a migrating LS member might have made between 1981 and 1991.

Given that analysis may not be undertaken at a spatial unit smaller than a district or borough (to maintain anonymity in the data), such units, or bigger, have only been used in previous research using the LS. However, using a new technique, the 'G' areas were 'built' from gentrified wards identified in the census analysis in order that a more selective geography of professionalisation could be used, instead of arbitrary political units. This method allowed for the retention of anonymity while using spatial units based on social rather than administrative or political characteristics.

The 'G' areas were labelled G1, G2, G3 and G4 ranging from 'high or ultra gentrification' to 'marginal gentrification' respectively and leaving the rest of London (ROL) as a unit in its own right. It was not now possible to sub-divide the ROL area to chart the start and destination locations of moves because wards had been unevenly extracted from the London boroughs to create the 'G' areas. A sixth area of the south-east (ROSE: Bedfordshire, Berkshire, Buckinghamshire, East and West Sussex, Essex, Hertfordshire, Kent, Oxfordshire and Surrey) and a seventh, the rest of England and Wales (ROEW), were added to examine the flows to and from the 'G' areas.

All of the variables selected were for LS members and so provide individual rather than household counts. An approach examining which part of a household these LS members formed was abandoned given the complex interplay between individual trajectories and household changes over time and in relation to the additional movement of LS members through space. However, this does mean that all figures given can be considered to be underestimates of the processes under examination.

Gentrifiers were measured using the same socioeconomic groups as for the census. The variable created was derived from those LS

members who were 10 years old and more in 1981. This meant that it would be possible to observe the changes for all LS members who would be of working age by the 1991 census and who had had time to express at least a limited degree of housing choice. The displaced variables used were also constructed in the same way as set out in the use of the census data. Ethnicity became problematic in the context of the LS since it is underrepresented and was therefore eliminated from the analysis. The final variables analysed were:

- (1) *Working class*: LS members who were 18 years old and above in 1991 were selected, plus all those who were in this occupational category in either 1991 or 1981.
- (2) *Unskilled labour*: The next most significant variable from the census analysis; the same groups as in the census data were used for all those LS members in this category in 1981 and 1991.
- (3) *Elderly*: All those LS members were taken who were aged 55 years or over for men in 1981 and 50 and over for women in 1981. This meant that pensionable age would have been reached by 1981 for both sexes.
- (4) *Lone parents*: Comprised of all those households that were lone-parented at 1981 and/or 1991 to analyse family structure alteration over that time-period and in relation to moves made from and to the 'G' areas.
- (5) *Unemployed*: All LS members at both time-points and in relation to the 'G' areas in this state.

Incumbent Upgrading or Gentrification?

Having manually 'built' four new borough-sized areas from gentrified wards, checks were made to see if what appeared to be gentrification was based on migration or incumbent upgrading; and, if an associated out-migration by those groups seen as possible displacees was greater than migration rates in the capital as a whole. This was done to demonstrate whether or not the 'G' areas had

Table 2. Social and tenure position for non-moving LS members in ROL area

Position for 1981	Position in 1991			
	Professionals		Working class	
	Owners	Renters	Owners	Renters
<i>Working class</i>				
Owners	1034	<i>16</i>	2313	18
Renters	29	158	49	157
<i>Professionals</i>				
Owners	4841	55	1399	<i>19</i>
Renters	95	292	27	58

Source: ONS Longitudinal Study.

been actively gentrified or whether social changes in these areas accounted for an image of gentrification mistakenly derived from the cross-sectional census data.

Table 2 sets the scene by describing changes in the capital between 1981 and 1991 by looking at those LS members who had not moved (figures for renting are for private renting). It should be remembered that all tables represent raw data figures and should be multiplied by 100 to obtain the ‘real figure’ represented by the sample figures.

Table 2 shows the occupational and tenure position of LS members across the ROL area for 1981 down the left-hand side and for 1991 in the top row. To track movements within or between these positions, one can see, for example, that 1034 LS members were working-class owners in 1981 but had become professional owners in 1991. This example estimates that a total of 103 400 people in London remained in the same location but shifted their occupational status.

Table 2 shows that in the ROL area those who did not move stayed largely in the same occupational category (though the data only apply to working-class and professional occupations). Interestingly, there was some net loss of professionals to working-class occupations. Moves between tenures on changing occupational class were very low (see numbers in italics in Table 2) as one would

expect given that these are the non-moving group. Only 29 LS members were working-class renters and became professional owners by 1991—indicating that this group was not a significant outlet for the working class. Table 3 describes the social and tenure changes of the non-moving LS members over the period, this time for the ‘G’ areas. Table 3 may be read in the same way as Table 2, but refers to working-class non-movers in the ‘G’ areas rather than in the ROL area. Moves are compared for those moving from working-class to professional status (incumbent upgrading) and those remaining working class. Numbers in bold indicate tenure shifts by SEG for the decade for non-movers in the ‘G’ areas.

While the most significant trend is of tenure consistency over the period, again as one might expect for non-movers, the most significant of the ‘upward’ trends was for working-class owners to become professional owners. Roughly twice as many working class remained so in the ‘G’ areas, suggesting that little upgrading had occurred above the London average (a total of 1034 LS members moving from working-class owning to professional owning and 2313 retaining both the same tenure and class across the ROL area). This shows that no more incumbent upgrading was occurring in the ‘G’ areas than in the wider London area over the decade.

Table 3. Working-class non-movers in 1981 and their position in 1991, by 'G' area

	G1	G2	G3	G4
<i>Upwardly mobile non-movers^a</i>				
Working-class owners to Professional owners	44	51	63	53
Working-class owners to Professional renters	1	1	0	0
Working-class renters to Professional owners	1	2	0	0
Working-class renters to Professional renters	17	11	14	16
<i>Occupationally static non-movers^b</i>				
Working-class owners to working-class owners	78	106	120	124
Working-class owners to working-class renters	3	1	0	0
Working-class renters to working-class owners	4	11	2	6
Working-class renters to working-class renters	4	19	8	14

Source: ONS Longitudinal Study.

^aMove from working class to professional.

^bRemain working class.

It is now possible to contrast this picture of occupational and tenure stasis with the LS *movers* to the 'G' areas to see if a gentrification process was more significant than the unremarkable occupational upgrading exhibited by the 'G' areas. If we compare Table 4 with Table 5, we can see that the raw numbers of working-class households moving to the 'G' areas by 1991 was massively exceeded by the numbers for professionals. Interestingly, the lower 'G' areas experienced larger migratory flows to them by both professionals and working-class groups and it may be that less extensively gentrified areas allowed easier housing market access to working-class groups and provided a better investment choice for professional groups.

The data also correspond with theories on the previous location of gentrifiers before

moving, since the majority of professional movers are already from the London area. Le Gates and Hartman (1986) note that the 'back to the city' theory was a misnomer and that roughly 64 per cent of in-moves were made within the city this compares favourably with the 46 per cent seen here. This may also be a source of some dismay for those who see the renaissance of inner London as a 'back to the city' movement, rather than a churning of the supply of higher-status households in the metropolitan area.

In aggregate, 249 working-class LS members moved to the 'G' areas, relatively insignificant when compared to the 1071 LS-member professionals moving to them. These moves are given greater significance if one looks at the total number of professional moves made to the Greater London area by professionals from ROSE and ROEW (a total

Table 4. Origin of LS professional movers migrating to the 'G' areas by 1991

Location in 1981	Location in 1991				Total
	G1	G2	G3	G4	
ROL	77	127	149	146	499
ROSE	58	65	44	96	263
ROEW	79	89	77	62	307
Total	214	281	270	306	1071

Source: ONS Longitudinal Study.

Table 5. Origin of LS working-class movers migrating to the 'G' areas by 1991

Location in 1981	Location in 1991				Total
	G1	G2	G3	G4	
ROL	23	40	65	55	183
ROSE	9	7	8	8	32
ROEW	5	5	6	9	25
Total	36	52	89	72	249

Source: ONS Longitudinal Study.

1324) which was only 255 more than the number of professional moves to the 'G' areas. Only 200 LS working-class members moved from ROSE and ROEW to the ROL area. The total number of professional moves *within* London came to 2851 showing that roughly one-third of all moves were made to 'G' areas. This provides evidence that the 'G' areas were a critical locus for gentrification over the decade.

As a final check to see that the 'G' areas had indeed been gentrified, Table 6 contrasts the number of new entrants to the professional group in the 'G' areas from working-class occupations with those professionals entering the 'G' areas—to measure the extent of gentrification in the 'G' areas. Table 6 shows that, although there was some movement to professional status by working-class LS members upon ending up in the 'G' areas by 1991, this was insignificant compared with those who were professionals at both census points. In a sense, this is less important than the preceding tables given that gentrification could still be considered to have occurred even if it was

through occupational mobility upon making moves to the 'G' areas.

To establish finally the real scale of the flows to the 'G' areas, a net figure was calculated. The total flow by professionals to the 'G' areas by 1991 was 1069, while 881 had moved to other areas, showing that a net increase of 188 had occurred in these areas. This represents a 20 per cent increase in the number of professionals. In fact, the total number of moves made by professionals (1952) to and from the 'G' areas formed more than two-thirds of the total number of migratory moves made by that occupational group in the metropolitan area (2851) over the decade.

Displacement from Gentrification

Following the verification of gentrification as an active process in the 'G' areas, it was hypothesised that flows of displacees from these areas would increase with levels of gentrification—although we have already seen that flows to the lower 'G' areas were more extensive and an adjustment of expec-

Table 6. Number of moves to the 'G' areas by LS members previously working-class or professional

Status in 1981 outside the 'G' areas	Professionals migrating to the 'G' areas by 1991			
	G1	G2	G3	G4
Professionals	214	281	270	304
Working class	63	65	77	69

Source: ONS Longitudinal Study.

Table 7. Flows of working class to and from the 'G' areas

From			To/From	To		
ROEW	ROSE	ROL		ROL	ROSE	ROEW
5	9	23	G1	60	19	13
5	7	40	G2	57	27	16
6	8	65	G3	52	30	30
9	8	55	G4	66	37	26

Source: ONS Longitudinal Study.

tations may necessarily be warranted. Each displacee group is examined in turn.

Working-class Movers

The number of working-class LS members moving over the period in the ROL area totalled 1217, while moves made by working-class members to and from the 'G' areas were 673 indicating that the 'G' areas formed a concentrated locus of migration activity for this occupational group. Though a crude technique, it is possible to give ward averages of migration by dividing the total number of moves in an area by the number of wards in it. Working-class moves were at the rate of 1.61 moves per ward in the ROL area whereas those in the 'G' areas were at 5.6 moves per ward.

The number of working-class moves within the 'G' areas was almost undetectable compared to the number made to the ROL, ROSE and ROEW areas. This suggests a far greater propensity for moves by this group to be made to areas other than the 'G' areas rather than remaining in them.

Table 7 (and subsequent LS data tables) allow us to examine the flows of migrating LS members by 'G' area. The columns to the left of the central column show the number of LS members who moved from areas outside the 'G' areas to them and their origin. The columns to the right show the destination of moves made by working-class LS members from the 'G' areas by the 1991 census. The number of working-class LS members who did not move over the period is included here.

The first thing that one notices is that the moves away from the 'G' areas clearly exceed those of the working-class moves to those areas. By subtracting the number of moves made away from the 'G' areas from those made to the 'G' areas, net losses for the 'G' areas were as follows: G1–55, G2–48, G3–33 and G4–57. A total loss of 193 LS members from the 'G' areas.

As mentioned earlier, it would seem that the higher 'G' areas did not displace more working-class groups. However, we saw that greater numbers of professionals flocked to the lower 'G' areas, so this result is compatible with that initial finding. One should also bear in mind that to characterise these flows as pure displacement or otherwise may be mistaken; the proportion of moves due to choice and those due to some form of displacement can only be speculated upon.

In total, flows of the working class in 1981 to the 'G' areas by 1991 were only 240 while the outflow was 433. This figure is an underestimate since it singles out those who did not change occupation on moving to or from the 'G' areas. Interestingly, the net loss to the 'G' areas approximates the figure for the net increase in professionals in the 'G' areas (188); although to attribute a simple causal relationship would be simplistic, this points to a displacement effect of some kind.

Given that the outflow of working-class LS members from the 'G' areas was much higher than overall levels of mobility by this group for the rest of the capital, it would seem unlikely that these moves can be characterised by choice alone (Lyons, 1995). The major destination for these moves was to the

Table 8. Flows of elderly (by 1991) in relation to the 'G' areas

From			To/From	To		
ROEW	ROSE	ROL		ROL	ROSE	ROEW
2	2	42	G1	44	31	27
4	7	58	G2	34	42	32
6	9	62	G3	57	55	39
6	4	62	G4	56	40	39

Source: ONS Longitudinal Study.

ROL area (235 of the 433 moves out of the 'G' areas) supporting the conclusion by Lyons that moves by those with fewer resources are characterised by constraint and tend to be over small distances.

Elderly Movers

A net loss of the elderly of 232 LS members was observed (264 entered the 'G' areas and 496 left). Without a socioeconomic breakdown, it is not possible to verify the characteristics of the elderly moving to the 'G' areas and those from them, although it might be plausible to hypothesise that movement to the areas was made by higher SEGs and that from the areas was by lower SEGs.

Table 8 shows these flows in more detail. Moves to the 'G' areas seem to cluster in the lower 'G' areas. Also of note, in contrast to that of the working-class group, is the greater dispersal of those leaving the 'G' areas to each of the destination areas. It may be that many moves were made that were either highly localised or to the exterior of the metropolitan area. In terms of migration to the 'G' areas by this group, such moves were mostly made by people already living in the Greater London area.

Why should this pattern be so evident? It may be that, if some of these moves are made by displacees, they move to friends and family; while others, utilising greater resources, move to places of retirement—although one cannot be sure of this from the data, it would seem plausible.

If we compare the losses to the 'G' areas with non-movers in them (approximately

1200), we see that there was an approximate net loss of 18 per cent from this group. Indeed, the loss from the 'G' areas compared with that for the ROL as a whole (200) shows that the 'G' areas' loss (232) was greater. This represents a staggering level of change in relation to an area (all of the 'G' areas together) roughly one-eighth the size of London. Henig (1984) has indicated that the costs to the elderly are profound in gentrified areas and it would seem that, from the above results, gentrification over the decade occurred in areas with a high proportion of elderly people and appeared to displace or replace a massive migratory flow of such people away from such areas.

Unskilled-labour Movers

The LS data revealed indistinct patterns of migration for LS members in unskilled labour, stemming in part from the relatively small proportion of the overall workforce that this group represents.

Table 9 shows that the flows to and from the 'G' areas by these groups were insignificant. There was a net loss of only 2 unskilled workers from the 'G' areas, in total. Levels of migration can be compared to the LS unskilled non-movers, of whom 23 remained in the 'G' areas over the period; these were evenly distributed across the 'G' areas. The figure for the rest of the London was 191. This means that the 'G' areas were holding a below-average number of unskilled workers over the period. This would appear to indicate that this group could not be associated with a migration of any significance

Table 9. Flows to and from the 'G' areas of unskilled LS members

From			To/From	To		
ROEW	ROSE	ROL		ROL	ROSE	ROEW
0	0	4	G1	3	0	0
0	0	1	G2	2	0	0
0	0	0	G3	0	0	1
0	0	0	G4	0	0	1

Source: ONS Longitudinal Study.

and one that would be difficult to connect to gentrification.

Unemployed Movers

In the case of working-class and other SEG measurements, the interaction between occupation and geography had been examined at both census points but unemployment was taken as a 'destination' measurement. Using a destination measure would include those who had become unemployed and eliminate those who had been unemployed but who found work by the time of the second census which might be difficult to view as displacement.

Table 10 shows that the net changes in the 'G' areas were again insignificant overall for this group. The total net change in the 'G' areas was an additional 2 LS members. 'G' areas 3 and 4 lost only 2 and 1 LS members respectively.

Inactive Movers

The inactive group was added in the LS analysis because it comprises the perma-

nently sick and disabled, housewives, the retired, students and persons of independent means; a potentially effective proxy for displacement. As with the unemployed, a destination measure was utilised—all those who were inactive by 1991.

Table 11 shows that, while movements to and from the 'G' areas and the ROL were roughly the same, the net losses for each 'G' area were G1–85, G2–97, G3–109 and G4–94 (a total net loss of the inactive from the 'G' areas of 385). These dramatic migratory losses are further highlighted when contextualised with moves across the city—given that only 261 LS inactive members moved in the capital over this period. As with the figure provided for the elderly, one can see a remarkably high movement away from the 'G' areas yet this figure exceeds the total losses to the 'G' areas by the working-class group. This indicates that the 'G' areas, again, represented a massive locus for migration wherein nearly one and a half times more moves were made away from the 'G' areas than within London as a whole in the same period.

It is worth noting at this stage that the

Table 10. Flows to and from the 'G' areas of LS members unemployed by 1991

From			To/From	To		
ROEW	ROSE	ROL		ROL	ROSE	ROEW
7	6	24	G1	29	5	8
10	4	35	G2	32	11	6
7	3	29	G3	21	9	7
5	3	22	G4	27	7	5

Source: ONS Longitudinal Study.

Table 11. Flows to and from the 'G' areas of LS members inactive by 1991

From			To/From	To		
ROEW	ROSE	ROL		ROL	ROSE	ROEW
11	9	97	G1	107	50	45
14	13	16	G2	111	72	57
18	18	114	G3	105	89	65
11	12	125	G4	117	74	51

Source: ONS Longitudinal Study.

unidimensional identification of displacee groups by different census categories provides problems insofar as they are not mutually exclusive. This means that one might be elderly and inactive, for example. This adds an extra consideration to be pondered when considering whether or not the flows singled out are to be interpreted as displacement. If one added all of the flows, one would exceed the input to the areas by professionals so it is clear that only a part of them could be seen as displacement.

Lone-parent Movers

In the ROL area, of those LS members who were lone parents at both census dates, 344 moved within that area and 1501 were non-movers. One may contrast these levels with that experienced in the 'G' areas to see if lone parents were more prone to move out of the 'G' areas than those who moved around London. Interestingly, lone parents showed an apparent 'liking' for the 'G' areas with a

net increase of 6 LS members, but this is an insignificant number. In relation to the ROL area, the moves to and from the 'G' areas are unremarkable so effectively ending any potential association with a displacement effect.

Summary of the Displacement Evidence

Table 12 summarises the key data from the previous LS-derived tables. Column 2 compares the net flows (gains and losses), as a total for all of the 'G' areas together. This figure is the LS figure multiplied from its 1 per cent root. This figure is then expressed, in column 3, as a percentage of the non-movers in the 'G' areas.

Table 12 shows that the overall gains to the 'G' areas by the professionals over the decade (18 000) will have been a significant displacing force. For whom this was most strongly felt one cannot say with certainty; it is unlikely that many more than this would be displaced. Losses of working-class, inactive and elderly groups appear profound.

Table 12. Net flows and percentage gain or loss for all 'G' areas between 1981 and 1991

Variable	Net change for all 'G' areas	Percentage gain/loss
Professionals	+ 18 800	+ 20
Inactive	- 38 500	- 46
Working class	- 19 300	- 38
Elderly	- 23 200	- 18
Unskilled	between - 200 and - 1 800	between - 9 and - 78
Unemployment	between - 1 600 and + 100	between - 4 and - 59
Lone parents	+ 600	+ 4.5

Source: ONS Longitudinal Study.

They are way above the metropolitan average and occur in areas which had been extensively professionalised suggesting that such social forces had an influence on these groups. This supports both an intuitive reading of the data presented and corresponds with past literature on gentrification and displacement and whom these processes touch.

As a summary, such information does little to help us to understand the qualitative impacts that migration has on people who make location decisions out of choice and coercion rather than free will—although the definition of such factors is, at best, contentious. This paper has attempted to provide a new reading of the displacement phenomenon by building a geography of gentrification and examining its relationship to other groups in the city. We have seen that groups such as the elderly have experienced profound flows away from professionalised locations in London, although we will never know whether this was because of rental increases, landlord harassment or the decision simply to move on. However, in combination with flows within London and the sparseness of flows to the 'G' areas, it would seem self-evident that such flows cannot be wholly due to choice or chance. It is likely that a displacement effect has been clustered in those areas where professionalisation has occurred.

Conclusion: Measuring the Invisible

This paper presents evidence suggesting that gentrification was an active and extensive process during the 1980s in Greater London and that, where this occurred, it was associated with above-average levels of losses of groups who have been associated with displacement in the past. Labour market changes, tenure preferences and housing policies may explain some of the changes examined, but the approach allows us to pinpoint a potential causal mechanism—gentrification represented by the proxy indicator of professionalisation. The difficulties of directly quantifying the amount of displacement and replacement and other 'noise' in the data are hard to overcome. It may be that

further research at a finer spatial scale using a more qualitative approach could usefully supplement this work.

Using a re-aggregated geography of ward-level change, the interaction between professionalisation and other migratory changes was more closely linked to the processes of gentrification and displacement. Using the arbitrary distinctions imposed by an analysis using local authority districts for social building-blocks blunts such analysis. The breakdown used in this study permits a closer tracking of displacees via the LS. This has shown that gentrification had occurred in the 'G' areas and that this was related to massive flows, in certain cases, away from such areas by those less well resourced and previously associated with a displacement effect.

The figures presented should not let us forget that displacement also implies a social cost, likely to be in the form of increased housing need, overcrowding in 'hidden households' and homelessness. This is quite apart from the resentment and feelings of disenfranchisement and exclusion that people are likely to feel where they find themselves priced-out of a dwelling to make way for a higher-paying household; harassed or financially induced to leave a property; or, unable to enter the residential market in an area where they have spent much of their lives or close to a workplace (Atkinson, 1997).

A number of implications stem from this work. It would seem clear that certain areas of London have undergone a social evacuation effect mediated by the process of professionalisation. This also means that the neighbourhood effects, in the form of impaired social networks and reduced service provision, adversely affect groups less able to cope with the psychological and financial costs of such shocks. Displacement removes social problems and rearranges rather than ameliorates the causes of poverty, environmental decay and the loss of neighbourhood vitality—problems are moved rather than solved. This paper has indicated that such processes have occurred and, although a recent census has not allowed a more contemporary

picture, it is likely that they have continued. As Hartman (1979, p. 23) has argued:

It is true that we do not have good estimates of the precise magnitude of the current displacement problem ... [but] those who claim that recent analogous history is not necessarily true of today's displacement problem must prove otherwise.

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